Functional Constraints
Flame competition constraint within pocket regions.
Penalty coefficient 0.0100
Maximum allowable heat flux density ratio (max/min) 1000.00
The linear burning rate of the first conditional fuel is greater than that of the second.
Penalty coefficient 0.0100
Constraint on maximum kinetic flame heat flux density.
Penalty coefficient 0.0100
Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09
Maximum kinetic flame heat flux density in skeleton structure 1.00e+08
Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08
Constraint on skeleton thickness to pore diameter ratio.
Penalty coefficient 0.0100
Threshold for skeleton thickness to pore diameter ratio 3.00
Constraint on skeleton thickness to large oxidizer particle diameter ratio.
Penalty coefficient 0.0100
Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints
Constraint on allowable surface temperature range
Minimum surface temperature 600 K
Maximum surface temperature 900 K

Differential Evolution Algorithm
Population size: 408
Mutation force: 0.5
Crossover probability: 0.9
Number of threads: 34
Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters
Shared Parameters (All Compositions)
Pre-exponential coefficient for gas phase reactions in inter-pocket medium 2421031.75342778
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas phase reactions in inter-pocket medium 50000.01714040933
Parameter bounds [5.000E+004; 2.500E+005]
Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 3581866.6844138955
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions in pocket region outside skeleton 50000.00271914318
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in inter-pocket medium 2.499999999816582
Parameter bounds [0.000E+000; 2.500E+000]
Reaction order for gas-phase chemistry outside skeleton structure 0.3032657545256975
Parameter bounds [0.000E+000; 2.500E+000]
Enthalpy difference between vaporization and binder decomposition 143547.11869004357
Parameter bounds [-1.000E+007; 1.000E+007]
Diffusion flame height correlation coefficient 0.0271
Parameter bounds [1.000E-003; 1.000E+001]
Power law exponent for burning rate dependency on A (skeleton thickness) 1.1490
Parameter bounds [0.000E+000; 3.000E+000]
Power law exponent for burning rate dependency on B (pore diameter) 0.5197
Parameter bounds [0.000E+000; 3.000E+000]
Radiation temperature correlation coefficient (average layer temperature) 0.0000
Parameter bounds [0.000E+000; 1.000E+000]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 9999999944958.76
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 186933.8456873988

Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 822839.2986421281
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 83726.69156359439
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.6591542385694311
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 1.5747354091406663E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0281477893989552E-12
Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9999996933653.494
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 136762.91900928
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 75804769655.00275
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 124350.78616073122
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.8928933304274367
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 1.2210086968383822E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0035343312057472E-12
Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9998575976201.34
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 145749.7484838436
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 108091958.5173069
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 63366.94421170975
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.0537236171259
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 2.7050071500651024E-07
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 9.81232422244999E-07
Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.0561
Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa
Experimental burning rate 16.94 mm/s
Calculated burning rate 9.32 mm/s
First Conditional Fuel
Surface temperature 647.84 K
Linear burning rate 10.68 mm/s
Mass decomposition rate 17.72 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.18e+07 W/m^2

Kinetic flame height 25.83 μm
Average kinetic flame density 1.82 kg/m^3
Average kinetic flame temperature 2,177.42 K
Binder sublimation heat flux density 1.18e+07 W/m^2
Surface heat balance error 0.0024 W/m^2

Secondary Conditional Fuel

Surface temperature 627.74 K
Linear burning rate 4.49 mm/s
Mass decomposition rate 7.45 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.02e+06 W/m^2
Kinetic flame height 86.1 μm
Average kinetic flame density 2.65 kg/m^3
Average kinetic flame temperature 1,496.37 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.56e+06 W/m^2
Kinetic flame height 17.33 μm
Average kinetic flame density 3.58 kg/m^3
Average kinetic flame temperature 1,109.37 K
Heat flux density due to metal combustion, 2.14e+06 W/m^2
Effective thermal conductivity 0.43 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.026 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.35e-04 m
Pore diameter 1.63e-05 m

Diffusion flame heat flux density 696,691.89 W/m^2
Diffusion flame height 427.2 μm
Heat flux density to surface within skeletal structure 2.77e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.29e+06 W/m^2
Total heat flux density to surface 4.76e+06 W/m^2
Binder sublimation heat flux density 4.76e+06 W/m^2
Surface heat balance error -0.0014 W/m^2

<u>*At pressure 4.06 MPa*</u>

Experimental burning rate 35.09 mm/s
Calculated burning rate 21.85 mm/s

First Conditional Fuel

Surface temperature 691.21 K
Linear burning rate 58.34 mm/s
Mass decomposition rate 96.79 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 7.1e+07 W/m^2
Kinetic flame height 4.25 μm
Average kinetic flame density 7.32 kg/m^3
Average kinetic flame temperature 2,199.1 K
Binder sublimation heat flux density 7.1e+07 W/m^2
Surface heat balance error -0.0078 W/m^2

Secondary Conditional Fuel

Surface temperature 645.16 K
Linear burning rate 9.55 mm/s
Mass decomposition rate 15.84 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.47e+06 W/m^2
Kinetic flame height 117.15 μm
Average kinetic flame density 10.69 kg/m^3
Average kinetic flame temperature 1,505.08 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.18e+07 W/m^2
Kinetic flame height 8.05 μm

Average kinetic flame density 14.4 kg/m^3
Average kinetic flame temperature 1,118.08 K
Heat flux density due to metal combustion, 4.87e+06 W/m^2
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.018 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5.67e-05 m
Pore diameter 1.1e-05 m
Diffusion flame heat flux density 4.05e+06 W/m^2
Diffusion flame height 73.09 µm
Heat flux density to surface within skeletal structure 5.49e+06 W/m^2
Heat flux density to surface outside skeletal structure 983,124.43 W/m^2
Total heat flux density to surface 1.05e+07 W/m^2
Binder sublimation heat flux density 1.05e+07 W/m^2
Surface heat balance error 0.0018 W/m^2

At pressure 6.5 MPa
Experimental burning rate 44.85 mm/s
Calculated burning rate 27.29 mm/s
First Conditional Fuel
Surface temperature 707.14 K
Linear burning rate 103.31 mm/s
Mass decomposition rate 171.39 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 1.3e+08 W/m^2
Kinetic flame height 2.31 µm
Average kinetic flame density 11.69 kg/m^3
Average kinetic flame temperature 2,207.07 K
Binder sublimation heat flux density 1.3e+08 W/m^2
Surface heat balance error 0.0066 W/m^2
Secondary Conditional Fuel
Surface temperature 649.99 K
Linear burning rate 11.68 mm/s
Mass decomposition rate 19.38 kg·s^{-1}·m^{-2}
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.39e+06 W/m^2
Kinetic flame height 123.51 µm
Average kinetic flame density 17.11 kg/m^3
Average kinetic flame temperature 1,507.49 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.59e+07 W/m^2
Kinetic flame height 5.92 µm
Average kinetic flame density 23.02 kg/m^3
Average kinetic flame temperature 1,120.49 K
Heat flux density due to metal combustion, 6.07e+06 W/m^2
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.016 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.49e-05 m
Pore diameter 9.91e-06 m
Diffusion flame heat flux density 6.03e+06 W/m^2
Diffusion flame height 48.95 µm
Heat flux density to surface within skeletal structure 5.97e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.01e+06 W/m^2
Total heat flux density to surface 1.3e+07 W/m^2
Binder sublimation heat flux density 1.3e+07 W/m^2
Surface heat balance error -0.004 W/m^2

Propellant Composition "Bas_3"

<u>*At pressure 1 MPa*</u>
Experimental burning rate 15.24 mm/s
Calculated burning rate 14.26 mm/s
First Conditional Fuel
Surface temperature 647.84 K
Linear burning rate 10.68 mm/s
Mass decomposition rate 17.72 $kg\cdot s^{-1}\cdot m^{-2}$
Kinetic flame heat flux density to surface 1.18e+07 W/m^2
Kinetic flame height 25.83 μm
Average kinetic flame density 1.82 kg/m^3
Average kinetic flame temperature 2,177.42 K
Binder sublimation heat flux density 1.18e+07 W/m^2
Surface heat balance error 0.0024 W/m^2
Secondary Conditional Fuel
Surface temperature 647.84 K
Linear burning rate 10.68 mm/s
Mass decomposition rate 17.72 $kg\cdot s^{-1}\cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 859,602.38 W/m^2
Kinetic flame height 199.76 μm
Average kinetic flame density 2.63 kg/m^3
Average kinetic flame temperature 1,506.42 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 2.9e+07 W/m^2
Kinetic flame height 6.81 μm
Average kinetic flame density 2.43 kg/m^3
Average kinetic flame temperature 1,635.42 K
Heat flux density due to metal combustion, 4.25e+06 W/m^2
Effective thermal conductivity 0.32 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.1 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 4.98e-05 m
Pore diameter 1.04e-05 m
Diffusion flame heat flux density 841,667.22 W/m^2
Diffusion flame height 359.54 μm
Heat flux density to surface within skeletal structure 1.04e+07 W/m^2
Heat flux density to surface outside skeletal structure 590,623.79 W/m^2
Total heat flux density to surface 1.18e+07 W/m^2
Binder sublimation heat flux density 1.18e+07 W/m^2
Surface heat balance error 0.0024 W/m^2

<u>*At pressure 4.06 MPa*</u>
Experimental burning rate 37.34 mm/s
Calculated burning rate 37.83 mm/s
First Conditional Fuel
Surface temperature 691.21 K
Linear burning rate 58.34 mm/s
Mass decomposition rate 96.79 $kg\cdot s^{-1}\cdot m^{-2}$
Kinetic flame heat flux density to surface 7.1e+07 W/m^2
Kinetic flame height 4.25 μm
Average kinetic flame density 7.32 kg/m^3
Average kinetic flame temperature 2,199.1 K
Binder sublimation heat flux density 7.1e+07 W/m^2
Surface heat balance error -0.0078 W/m^2
Secondary Conditional Fuel
Surface temperature 663.72 K

Linear burning rate 20.41 mm/s
Mass decomposition rate 33.87 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 694,747.1 W/m^2
Kinetic flame height 244.88 μm
Average kinetic flame density 10.63 kg/m^3
Average kinetic flame temperature 1,514.36 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 6.7e+07 W/m^2
Kinetic flame height 2.92 μm
Average kinetic flame density 9.79 kg/m^3
Average kinetic flame temperature 1,643.36 K
Heat flux density due to metal combustion, 7.94e+06 W/m^2
Effective thermal conductivity 0.3 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.073 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 2.37e-05 m
Pore diameter 7.41e-06 m
Diffusion flame heat flux density 5.43e+06 W/m^2
Diffusion flame height 55.49 μm
Heat flux density to surface within skeletal structure 1.75e+07 W/m^2
Heat flux density to surface outside skeletal structure 532,745.81 W/m^2
Total heat flux density to surface 2.34e+07 W/m^2
Binder sublimation heat flux density 2.34e+07 W/m^2
Surface heat balance error 0.0045 W/m^2

At pressure 6.5 MPa
Experimental burning rate 50.5 mm/s
Calculated burning rate 45.31 mm/s
First Conditional Fuel
Surface temperature 707.14 K
Linear burning rate 103.31 mm/s
Mass decomposition rate 171.39 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.3e+08 W/m^2
Kinetic flame height 2.31 μm
Average kinetic flame density 11.69 kg/m^3
Average kinetic flame temperature 2,207.07 K
Binder sublimation heat flux density 1.3e+08 W/m^2
Surface heat balance error 0.0066 W/m^2
Secondary Conditional Fuel
Surface temperature 666.21 K
Linear burning rate 22.53 mm/s
Mass decomposition rate 37.37 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 727,531.03 W/m^2
Kinetic flame height 233.5 μm
Average kinetic flame density 17.02 kg/m^3
Average kinetic flame temperature 1,515.6 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1e+08 W/m^2
Kinetic flame height 1.96 μm
Average kinetic flame density 15.69 kg/m^3
Average kinetic flame temperature 1,644.6 K
Heat flux density due to metal combustion, 8.74e+06 W/m^2
Effective thermal conductivity 0.29 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.07 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 2.11e-05 m
Pore diameter 7.04e-06 m
Diffusion flame heat flux density 8.93e+06 W/m²
Diffusion flame height 33.68 μm
Heat flux density to surface within skeletal structure 1.65e+07 W/m²
Heat flux density to surface outside skeletal structure 617,435.92 W/m²
Total heat flux density to surface 2.6e+07 W/m²
Binder sublimation heat flux density 2.6e+07 W/m²
Surface heat balance error -0.0087 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa
Experimental burning rate 9.73 mm/s
Calculated burning rate 10.04 mm/s
First Conditional Fuel
Surface temperature 647.84 K
Linear burning rate 10.68 mm/s
Mass decomposition rate 17.72 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.18e+07 W/m²
Kinetic flame height 25.83 μm
Average kinetic flame density 1.82 kg/m³
Average kinetic flame temperature 2,177.42 K
Binder sublimation heat flux density 1.18e+07 W/m²
Surface heat balance error 0.0024 W/m²
Secondary Conditional Fuel
Surface temperature 622.13 K
Linear burning rate 3.49 mm/s
Mass decomposition rate 5.79 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 30,293.2 W/m²
Kinetic flame height 1,386.03 μm
Average kinetic flame density 4.77 kg/m³
Average kinetic flame temperature 832.06 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.49e+06 W/m²
Kinetic flame height 41.57 μm
Average kinetic flame density 4.26 kg/m³
Average kinetic flame temperature 931.56 K
Heat flux density due to metal combustion, 5.35e+06 W/m²
Effective thermal conductivity 1.42 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.011 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 1.8e-04 m
Pore diameter 1.86e-05 m
Diffusion flame heat flux density 15,285.56 W/m²
Diffusion flame height 19,350.76 μm
Heat flux density to surface within skeletal structure 3.62e+06 W/m²
Heat flux density to surface outside skeletal structure 14,275.14 W/m²
Total heat flux density to surface 3.65e+06 W/m²
Binder sublimation heat flux density 3.65e+06 W/m²
Surface heat balance error 2.75e-04 W/m²

At pressure 4.06 MPa
Experimental burning rate 25.92 mm/s
Calculated burning rate 26.1 mm/s
First Conditional Fuel
Surface temperature 691.21 K

Linear burning rate 58.34 mm/s
Mass decomposition rate 96.79 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 7.1e+07 W/m^2
Kinetic flame height 4.25 μm
Average kinetic flame density 7.32 kg/m^3
Average kinetic flame temperature 2,199.1 K
Binder sublimation heat flux density 7.1e+07 W/m^2
Surface heat balance error -0.0078 W/m^2

Secondary Conditional Fuel

Surface temperature 640.99 K
Linear burning rate 8 mm/s
Mass decomposition rate 13.27 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 20,863.03 W/m^2
Kinetic flame height 1,922.1 μm
Average kinetic flame density 19.13 kg/m^3
Average kinetic flame temperature 841.5 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.96e+06 W/m^2
Kinetic flame height 20.29 μm
Average kinetic flame density 17.11 kg/m^3
Average kinetic flame temperature 941 K
Heat flux density due to metal combustion, 1.35e+07 W/m^2
Effective thermal conductivity 1.42 $W/(m \cdot K)$
Conductive thermal conductivity 1.41 $W/(m \cdot K)$
Radiative thermal conductivity 0.0074 $W/(m \cdot K)$
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 6.94e-05 m
Pore diameter 1.21e-05 m

Diffusion flame heat flux density 106,273.03 W/m^2
Diffusion flame height 2,765.53 μm
Heat flux density to surface within skeletal structure 8.62e+06 W/m^2
Heat flux density to surface outside skeletal structure 9,910.36 W/m^2
Total heat flux density to surface 8.73e+06 W/m^2
Binder sublimation heat flux density 8.73e+06 W/m^2
Surface heat balance error 5.24e-04 W/m^2

<u>*At pressure 6.5 MPa*</u>

Experimental burning rate 36.06 mm/s
Calculated burning rate 35.42 mm/s

First Conditional Fuel

Surface temperature 707.14 K
Linear burning rate 103.31 mm/s
Mass decomposition rate 171.39 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.3e+08 W/m^2
Kinetic flame height 2.31 μm
Average kinetic flame density 11.69 kg/m^3
Average kinetic flame temperature 2,207.07 K
Binder sublimation heat flux density 1.3e+08 W/m^2
Surface heat balance error 0.0066 W/m^2

Secondary Conditional Fuel

Surface temperature 647.63 K
Linear burning rate 10.59 mm/s
Mass decomposition rate 17.57 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 18,372.48 W/m^2
Kinetic flame height 2,146.52 μm
Average kinetic flame density 30.54 kg/m^3
Average kinetic flame temperature 844.82 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.72e+06 W/m²

Kinetic flame height 15.93 µm

Average kinetic flame density 27.32 kg/m³

Average kinetic flame temperature 944.32 K

Heat flux density due to metal combustion, 1.84e+07 W/m²

Effective thermal conductivity 1.42 W/(m·K)

Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 0.0064 W/(m·K)

Conductive thermal conductivity balance error 9.240271880983641E-10

Skeleton layer thickness 5.03e-05 m

Pore diameter 1.04e-05 m

Diffusion flame heat flux density 197,471.29 W/m²

Diffusion flame height 1,484.96 µm

Heat flux density to surface within skeletal structure 1.15e+07 W/m²

Heat flux density to surface outside skeletal structure 8,786.42 W/m²

Total heat flux density to surface 1.17e+07 W/m²

Binder sublimation heat flux density 1.17e+07 W/m²

Surface heat balance error 1.05e-04 W/m²

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s

Calculated burning rate 14.26 mm/s

First Conditional Fuel

Surface temperature 609.03 K

Linear burning rate 11.24 mm/s

Mass decomposition rate 18.64 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.14e+07 W/m²

Kinetic flame height 27.24 µm

Average kinetic flame density 1.84 kg/m³

Average kinetic flame temperature 2,158.02 K

Binder sublimation heat flux density 1.14e+07 W/m²

Surface heat balance error 0.0026 W/m²

Secondary Conditional Fuel

Surface temperature 600 K

Linear burning rate 7.48 mm/s

Mass decomposition rate 12.42 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.19e+06 W/m²

Kinetic flame height 148.53 µm

Average kinetic flame density 2.68 kg/m³

Average kinetic flame temperature 1,482.5 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 8.15e+06 W/m²

Kinetic flame height 12.16 µm

Average kinetic flame density 3.62 kg/m³

Average kinetic flame temperature 1,095.5 K

Heat flux density due to metal combustion, 8.35e+06 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.06e-08 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 3.38e-05 m

Pore diameter 1.28e-11 m

Diffusion flame heat flux density 430,755.56 W/m²

Diffusion flame height 711.77 µm

Heat flux density to surface within skeletal structure 6.24e+06 W/m²

Heat flux density to surface outside skeletal structure 739,047.91 W/m^2
Total heat flux density to surface 7.41e+06 W/m^2
Binder sublimation heat flux density 7.41e+06 W/m^2
Surface heat balance error -4.95e-04 W/m^2

At pressure 4.06 MPa
Experimental burning rate 38.01 mm/s
Calculated burning rate 37.93 mm/s
First Conditional Fuel
Surface temperature 649.86 K
Linear burning rate 61.32 mm/s
Mass decomposition rate 101.74 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 6.83e+07 W/m^2
Kinetic flame height 4.48 μm
Average kinetic flame density 7.39 kg/m^3
Average kinetic flame temperature 2,178.43 K
Binder sublimation heat flux density 6.83e+07 W/m^2
Surface heat balance error 0.0094 W/m^2
Secondary Conditional Fuel
Surface temperature 619.01 K
Linear burning rate 17.37 mm/s
Mass decomposition rate 28.81 kg·s^{-1}·m^{-2}
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 793,232.9 W/m^2
Kinetic flame height 220.11 μm
Average kinetic flame density 10.79 kg/m^3
Average kinetic flame temperature 1,492 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 5.39e+07 W/m^2
Kinetic flame height 1.8 μm
Average kinetic flame density 14.57 kg/m^3
Average kinetic flame temperature 1,105 K
Heat flux density due to metal combustion, 2.14e+07 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 1.33e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.29e-05 m
Pore diameter 8.25e-12 m
Diffusion flame heat flux density 2.29e+06 W/m^2
Diffusion flame height 132.97 μm
Heat flux density to surface within skeletal structure 1.51e+07 W/m^2
Heat flux density to surface outside skeletal structure 634,351.34 W/m^2
Total heat flux density to surface 1.8e+07 W/m^2
Binder sublimation heat flux density 1.8e+07 W/m^2
Surface heat balance error -0.64 W/m^2

At pressure 6.5 MPa
Experimental burning rate 52.88 mm/s
Calculated burning rate 52.83 mm/s
First Conditional Fuel
Surface temperature 664.86 K
Linear burning rate 108.56 mm/s
Mass decomposition rate 180.1 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 1.25e+08 W/m^2
Kinetic flame height 2.43 μm
Average kinetic flame density 11.8 kg/m^3
Average kinetic flame temperature 2,185.93 K
Binder sublimation heat flux density 1.25e+08 W/m^2

Surface heat balance error -0.036 W/m²
Secondary Conditional Fuel
Surface temperature 626.22 K
Linear burning rate 23.58 mm/s
Mass decomposition rate 39.12 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 677,317.95 W/m²
Kinetic flame height 256.72 µm
Average kinetic flame density 17.25 kg/m³
Average kinetic flame temperature 1,495.61 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1e+08 W/m²
Kinetic flame height 0.96 µm
Average kinetic flame density 23.27 kg/m³
Average kinetic flame temperature 1,108.61 K
Heat flux density due to metal combustion, 3e+07 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 1.13e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 9.05e-06 m
Pore diameter 7.04e-12 m
Diffusion flame heat flux density 3.08e+06 W/m²
Diffusion flame height 98.8 µm
Heat flux density to surface within skeletal structure 2.12e+07 W/m²
Heat flux density to surface outside skeletal structure 566,736.05 W/m²
Total heat flux density to surface 2.49e+07 W/m²
Binder sublimation heat flux density 2.49e+07 W/m²
Surface heat balance error 0.0046 W/m²

Propellant Composition "Bas_0"

At pressure 1 MPa
Experimental burning rate 5.97 mm/s
Calculated burning rate 6.04 mm/s
First Conditional Fuel
Surface temperature 825.01 K
Linear burning rate 8.81 mm/s
Mass decomposition rate 14.61 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.36e+07 W/m²
Kinetic flame height 21.12 µm
Average kinetic flame density 1.75 kg/m³
Average kinetic flame temperature 2,266 K
Binder sublimation heat flux density 1.36e+07 W/m²
Surface heat balance error 0.0038 W/m²
Secondary Conditional Fuel
Surface temperature 791.72 K
Linear burning rate 2.8 mm/s
Mass decomposition rate 4.65 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 3.55e+06 W/m²
Kinetic flame height 44.28 µm
Average kinetic flame density 2.51 kg/m³
Average kinetic flame temperature 1,578.36 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 6,676.13 W/m²
Kinetic flame height 11,972.16 µm
Average kinetic flame density 3.33 kg/m³
Average kinetic flame temperature 1,191.36 K

Heat flux density due to metal combustion, 1.52e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 3.51e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.35e-04 m
Pore diameter 2.18e-11 m
Diffusion flame heat flux density 1.08e+06 W/m²
Diffusion flame height 266.37 μm
Heat flux density to surface within skeletal structure 394,667.01 W/m²
Heat flux density to surface outside skeletal structure 2.63e+06 W/m²
Total heat flux density to surface 4.11e+06 W/m²
Binder sublimation heat flux density 4.11e+06 W/m²
Surface heat balance error 0.0012 W/m²

At pressure 4.06 MPa
Experimental burning rate 13.63 mm/s
Calculated burning rate 13.81 mm/s
First Conditional Fuel
Surface temperature 879.86 K
Linear burning rate 48.16 mm/s
Mass decomposition rate 79.91 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 8.12e+07 W/m²
Kinetic flame height 3.48 μm
Average kinetic flame density 7.02 kg/m³
Average kinetic flame temperature 2,293.43 K
Binder sublimation heat flux density 8.12e+07 W/m²
Surface heat balance error 0.0077 W/m²
Secondary Conditional Fuel
Surface temperature 813.24 K
Linear burning rate 5.94 mm/s
Mass decomposition rate 9.85 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.59e+06 W/m²
Kinetic flame height 59.96 μm
Average kinetic flame density 10.13 kg/m³
Average kinetic flame temperature 1,589.12 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 8,268.22 W/m²
Kinetic flame height 9,406.6 μm
Average kinetic flame density 13.39 kg/m³
Average kinetic flame temperature 1,202.12 K
Heat flux density due to metal combustion, 3.45e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.38e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5.69e-05 m
Pore diameter 1.48e-11 m
Diffusion flame heat flux density 6.28e+06 W/m²
Diffusion flame height 45.46 μm
Heat flux density to surface within skeletal structure 648,742.4 W/m²
Heat flux density to surface outside skeletal structure 2.1e+06 W/m²
Total heat flux density to surface 9.03e+06 W/m²
Binder sublimation heat flux density 9.03e+06 W/m²
Surface heat balance error -0.0014 W/m²

At pressure 6.5 MPa
Experimental burning rate 18 mm/s

Calculated burning rate 17.82 mm/s

First Conditional Fuel

- Surface temperature 900 K
- Linear burning rate 85.31 mm/s
- Mass decomposition rate 141.53 $kg \cdot s^{-1} \cdot m^{-2}$
- Kinetic flame heat flux density to surface 1.48e+08 W/m^2
- Kinetic flame height 1.9 µm
- Average kinetic flame density 11.2 kg/m^3
- Average kinetic flame temperature 2,303.5 K
- Binder sublimation heat flux density 1.48e+08 W/m^2
- Surface heat balance error -0.023 W/m^2

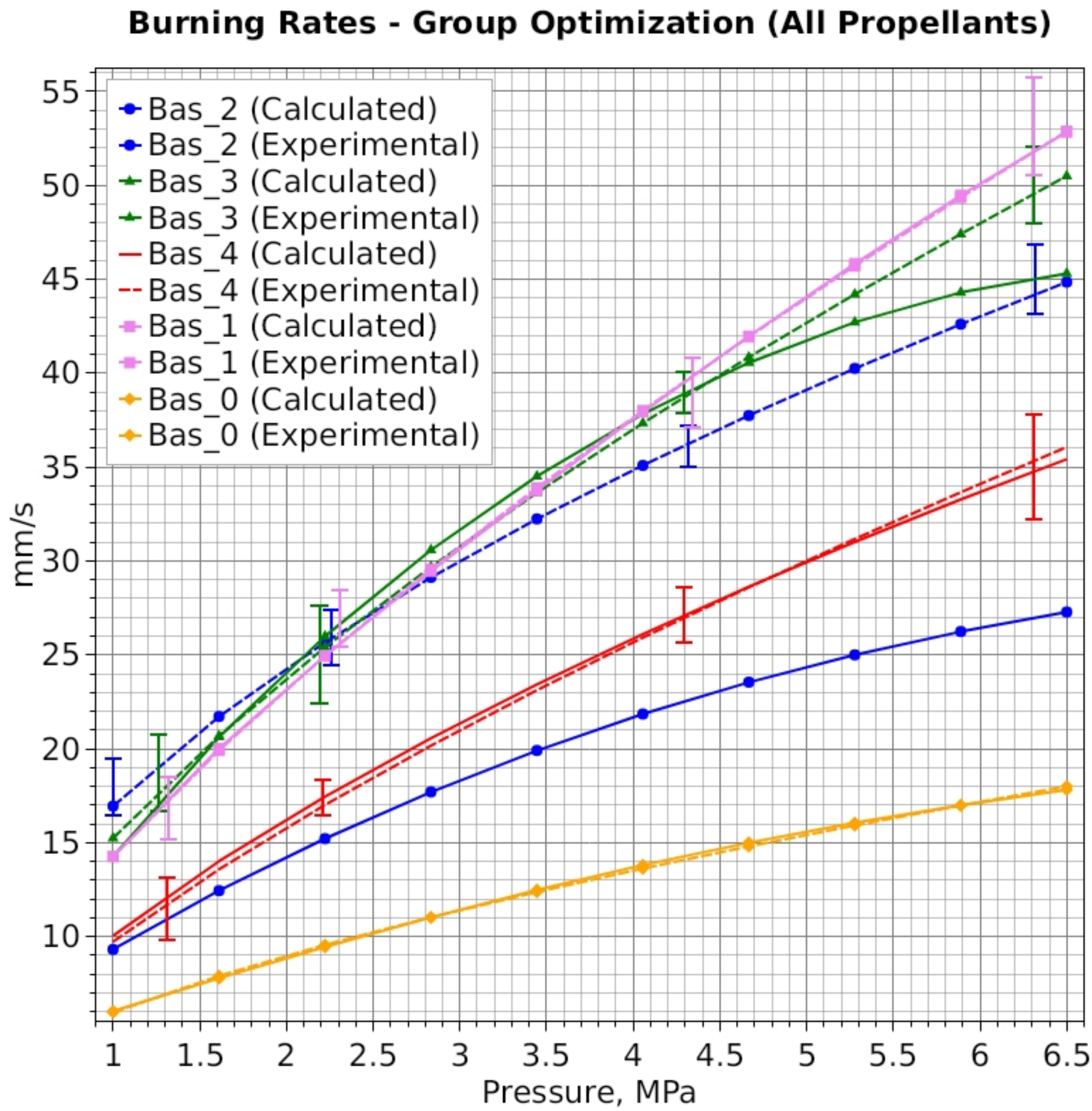
Secondary Conditional Fuel

- Surface temperature 820.38 K
- Linear burning rate 7.55 mm/s
- Mass decomposition rate 12.53 $kg \cdot s^{-1} \cdot m^{-2}$
- *Region Outside Skeleton Structure*
  - Kinetic flame heat flux density to surface 2.35e+06 W/m^2
  - Kinetic flame height 65.59 µm
  - Average kinetic flame density 16.2 kg/m^3
  - Average kinetic flame temperature 1,592.69 K
- *Skeleton Structure Region*
  - Kinetic flame heat flux density to surface 8,992.22 W/m^2
  - Kinetic flame height 8,569.83 µm
  - Average kinetic flame density 21.4 kg/m^3
  - Average kinetic flame temperature 1,205.69 K
  - Heat flux density due to metal combustion, 4.48e+06 W/m^2
  - Effective thermal conductivity 0.4 $W/(m \cdot K)$
  - Conductive thermal conductivity 0.4 $W/(m \cdot K)$
  - Radiative thermal conductivity 2.1e-08 $W/(m \cdot K)$
  - Conductive thermal conductivity balance error -4.092294297874943E-08
  - Skeleton layer thickness 4.32e-05 m
  - Pore diameter 1.3e-11 m
- Diffusion flame heat flux density 8.99e+06 W/m^2
- Diffusion flame height 31.65 µm
- Heat flux density to surface within skeletal structure 566,333.04 W/m^2
- Heat flux density to surface outside skeletal structure 2.06e+06 W/m^2
- Total heat flux density to surface 1.16e+07 W/m^2
- Binder sublimation heat flux density 1.16e+07 W/m^2
- Surface heat balance error -0.0024 W/m^2

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	9.32 mm/s		14.26 mm/s		10.04 mm/s		14.26 mm/s		6.04 mm/s	
Difference	7.62 mm/s		0.98 mm/s		-0.31 mm/s		0.0069 mm/s		-0.07 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	10.68 mm/s	4.49 mm/s	10.68 mm/s	10.68 mm/s	10.68 mm/s	3.49 mm/s	11.24 mm/s	7.48 mm/s	8.81 mm/s	2.8 mm/s
Surface Temperature	647.84 K	627.74 K	647.84 K	647.84 K	647.84 K	622.13 K	609.03 K	600 K	825.01 K	791.72 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.18e+07 W/m^2		1.18e+07 W/m^2		1.18e+07 W/m^2		1.14e+07 W/m^2		1.36e+07 W/m^2	
Kinetic Flame Heat Flux Density (Skeleton)	5.56e+06 W/m^2		2.9e+07 W/m^2		1.49e+06 W/m^2		8.15e+06 W/m^2		6,676.13 W/m^2	
Kinetic Flame Height (Skeleton)	17.33 μm		6.81 μm		41.57 μm		12.16 μm		11,972.16 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.02e+06 W/m^2		859,602.38 W/m^2		30,293.2 W/m^2		1.19e+06 W/m^2		3.55e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	86.1 μm		199.76 μm		1,386.03 μm		148.53 μm		44.28 μm	
Diffusion Flame Heat Flux Density	696,691.89 W/m^2		841,667.22 W/m^2		15,285.56 W/m^2		430,755.56 W/m^2		1.08e+06 W/m^2	
Diffusion Flame Height	427.2 μm		359.54 μm		19,350.76 μm		711.77 μm		266.37 μm	
Metal Combustion Heat Flux Density	2.14e+06 W/m^2		4.25e+06 W/m^2		5.35e+06 W/m^2		8.35e+06 W/m^2		1.52e+06 W/m^2	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	21.85 mm/s		37.83 mm/s		26.1 mm/s		37.93 mm/s		13.81 mm/s	
Difference	13.24 mm/s		-0.49 mm/s		-0.18 mm/s		0.083 mm/s		-0.18 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	58.34 mm/s	9.55 mm/s	58.34 mm/s	20.41 mm/s	58.34 mm/s	8 mm/s	61.32 mm/s	17.37 mm/s	48.16 mm/s	5.94 mm/s
Surface Temperature	691.21 K	645.16 K	691.21 K	663.72 K	691.21 K	640.99 K	649.86 K	619.01 K	879.86 K	813.24 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	7.1e+07 W/m²		7.1e+07 W/m²		7.1e+07 W/m²		6.83e+07 W/m²		8.12e+07 W/m²	
Kinetic Flame Heat Flux Density (Skeleton)	1.18e+07 W/m²		6.7e+07 W/m²		2.96e+06 W/m²		5.39e+07 W/m²		8,268.22 W/m²	
Kinetic Flame Height (Skeleton)	8.05 µm		2.92 µm		20.29 µm		1.8 µm		9,406.6 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.47e+06 W/m²		694,747.1 W/m²		20,863.03 W/m²		793,232.9 W/m²		2.59e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	117.15 µm		244.88 µm		1,922.1 µm		220.11 µm		59.96 µm	
Diffusion Flame Heat Flux Density	4.05e+06 W/m²		5.43e+06 W/m²		106,273.03 W/m²		2.29e+06 W/m²		6.28e+06 W/m²	
Diffusion Flame Height	73.09 µm		55.49 µm		2,765.53 µm		132.97 µm		45.46 µm	
Metal Combustion Heat Flux Density	4.87e+06 W/m²		7.94e+06 W/m²		1.35e+07 W/m²		2.14e+07 W/m²		3.45e+06 W/m²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	27.29 mm/s		45.31 mm/s		35.42 mm/s		52.83 mm/s		17.82 mm/s	
Difference	17.56 mm/s		5.19 mm/s		0.64 mm/s		0.052 mm/s		0.18 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	103.31 mm/s	11.68 mm/s	103.31 mm/s	22.53 mm/s	103.31 mm/s	10.59 mm/s	108.56 mm/s	23.58 mm/s	85.31 mm/s	7.55 mm/s
Surface Temperature	707.14 K	649.99 K	707.14 K	666.21 K	707.14 K	647.63 K	664.86 K	626.22 K	900 K	820.38 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.3e+08 W/m²		1.3e+08 W/m²		1.3e+08 W/m²		1.25e+08 W/m²		1.48e+08 W/m²	
Kinetic Flame Heat Flux Density (Skeleton)	1.59e+07 W/m²		1e+08 W/m²		3.72e+06 W/m²		1e+08 W/m²		8,992.22 W/m²	
Kinetic Flame Height (Skeleton)	5.92 μm		1.96 μm		15.93 μm		0.96 μm		8,569.83 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.39e+06 W/m²		727,531.03 W/m²		18,372.48 W/m²		677,317.95 W/m²		2.35e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	123.51 μm		233.5 μm		2,146.52 μm		256.72 μm		65.59 μm	
Diffusion Flame Heat Flux Density	6.03e+06 W/m²		8.93e+06 W/m²		197,471.29 W/m²		3.08e+06 W/m²		8.99e+06 W/m²	
Diffusion Flame Height	48.95 μm		33.68 μm		1,484.96 μm		98.8 μm		31.65 μm	
Metal Combustion Heat Flux Density	6.07e+06 W/m²		8.74e+06 W/m²		1.84e+07 W/m²		3e+07 W/m²		4.48e+06 W/m²	



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: U = A · p^v, A = 1.285341E-05; v = 0.52.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: U = A · p^v, A = 2.203002E-06; v = 0.64.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: U = A · p^v, A = 6.137894E-07; v = 0.7.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: U = A · p^v, A = 9.000448E-07; v = 0.7.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.
Vieille's Power Law: U = A · p^v, A = 1.720582E-06; v = 0.59.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

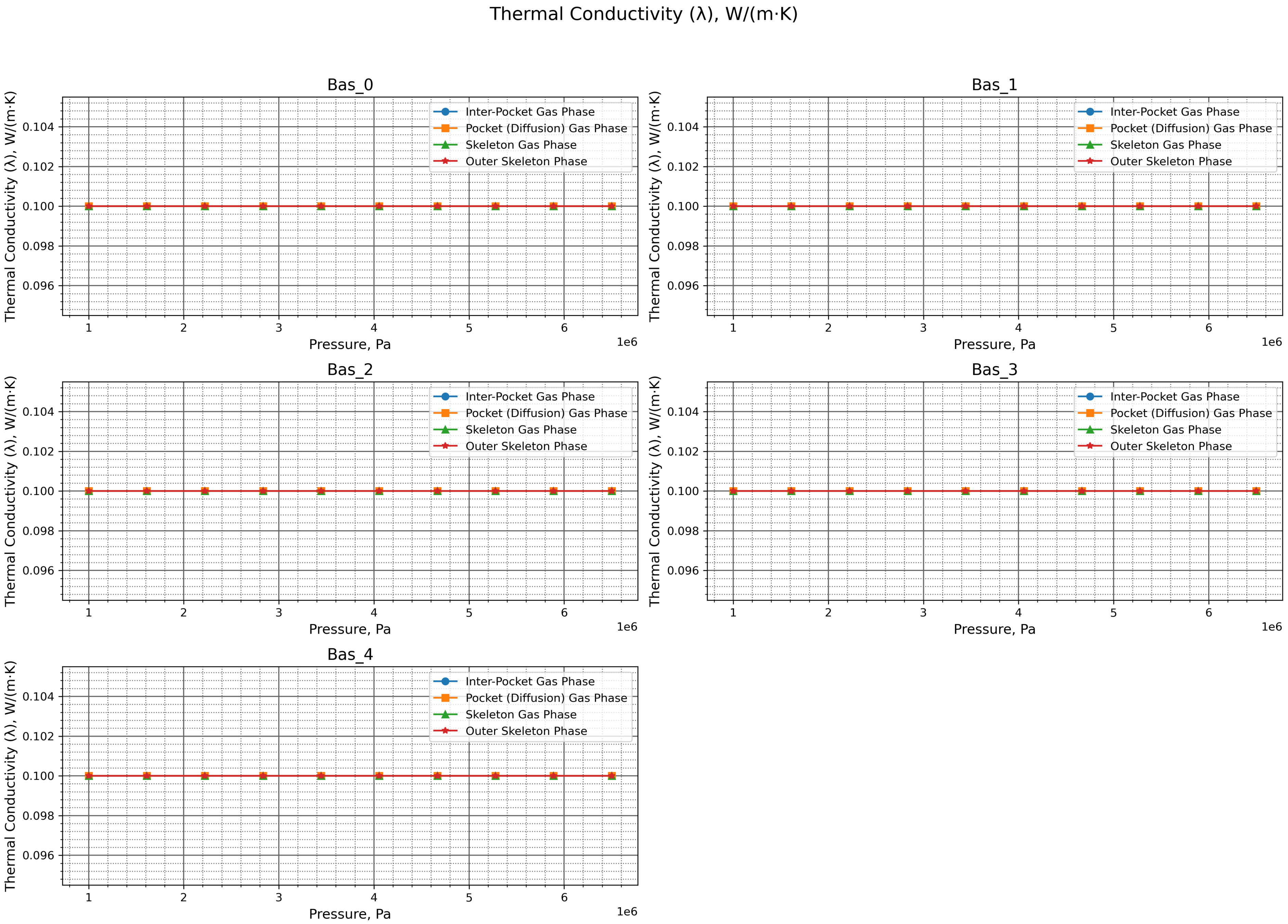
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,666 K

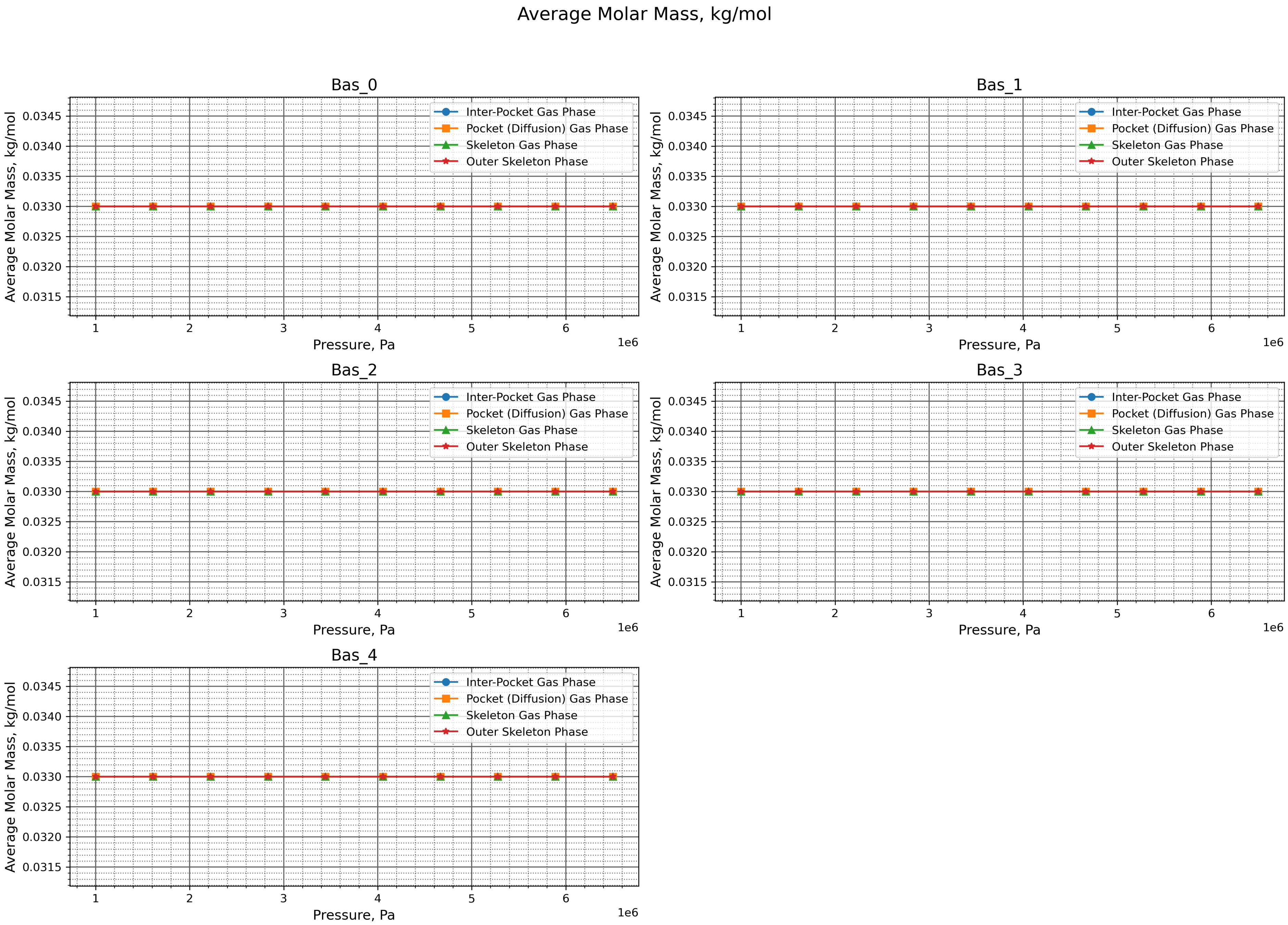
Gas Phase Parameters in Skeleton Structure

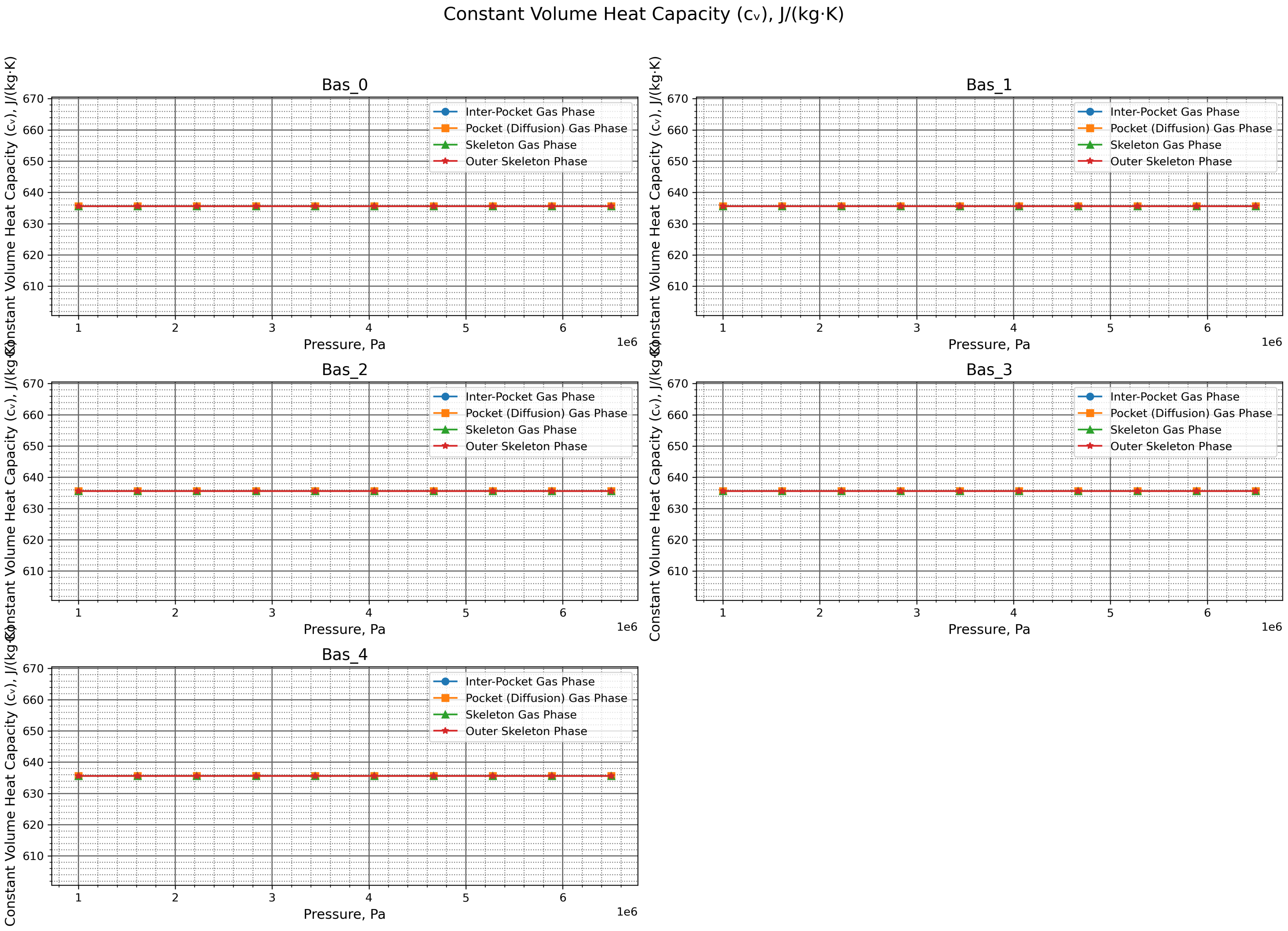
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

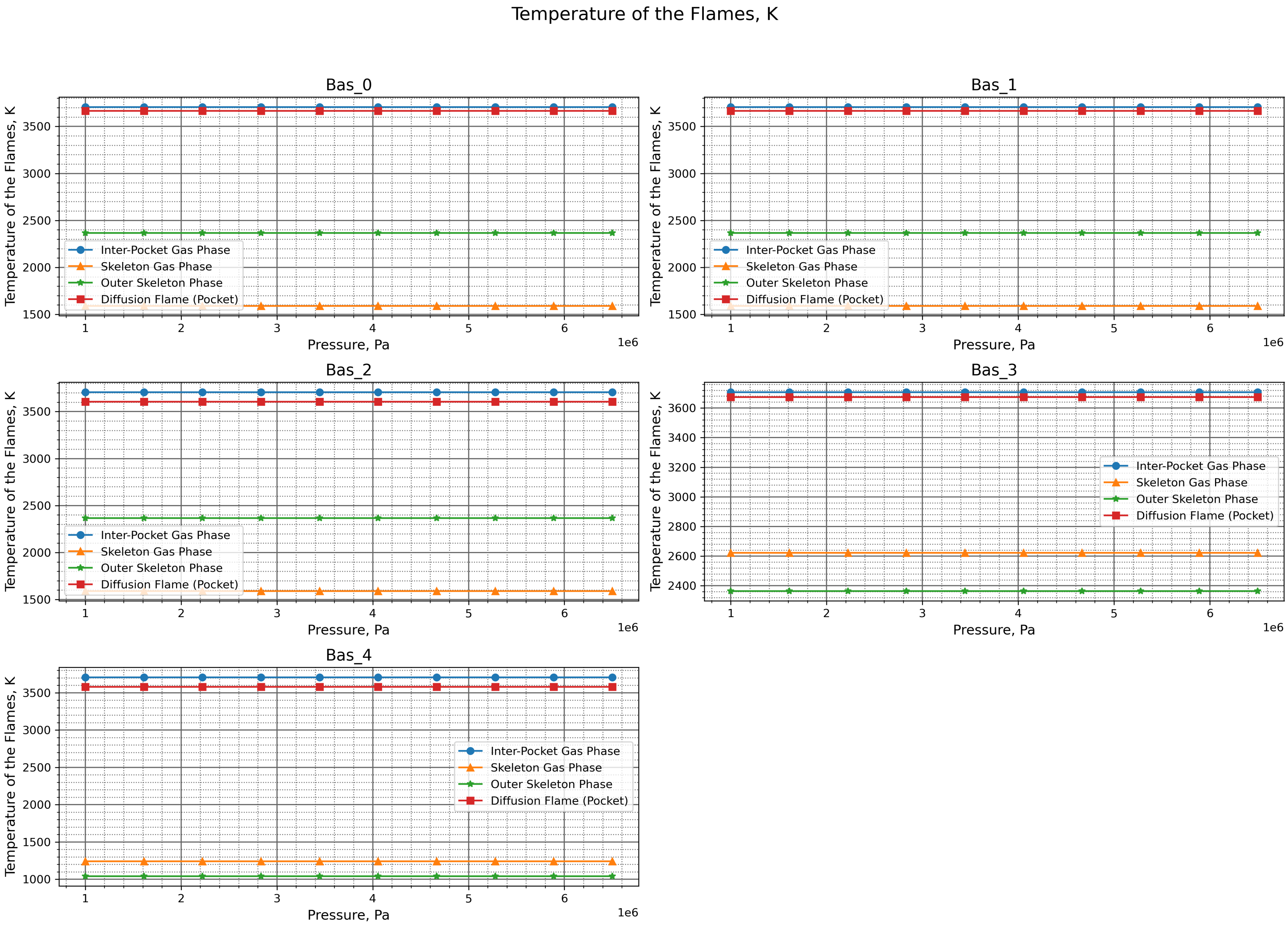
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

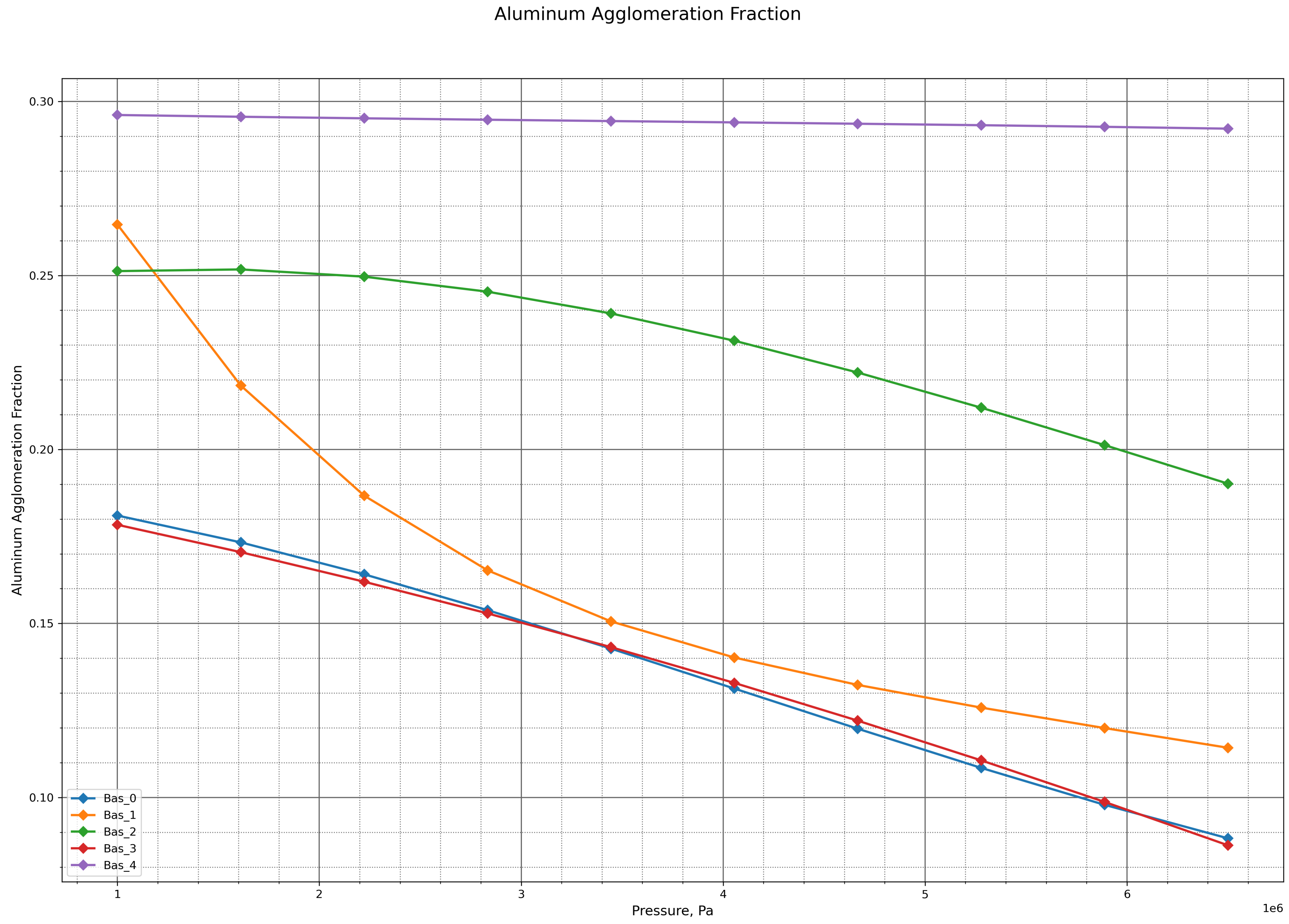

Thermal Conductivity (λ), W/(m·K)
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Thermal Conductivity (λ), W/(m·K)
Pressure, Pa
1e6
0.096
0.098
0.100
0.102
0.104
1
2
3
4
5
6


Average Molar Mass, kg/mol
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Average Molar Mass, kg/mol
0.0345
0.0340
0.0335
0.0330
0.0325
0.0320
0.0315
Pressure, Pa
1
2
3
4
5
6
1e6

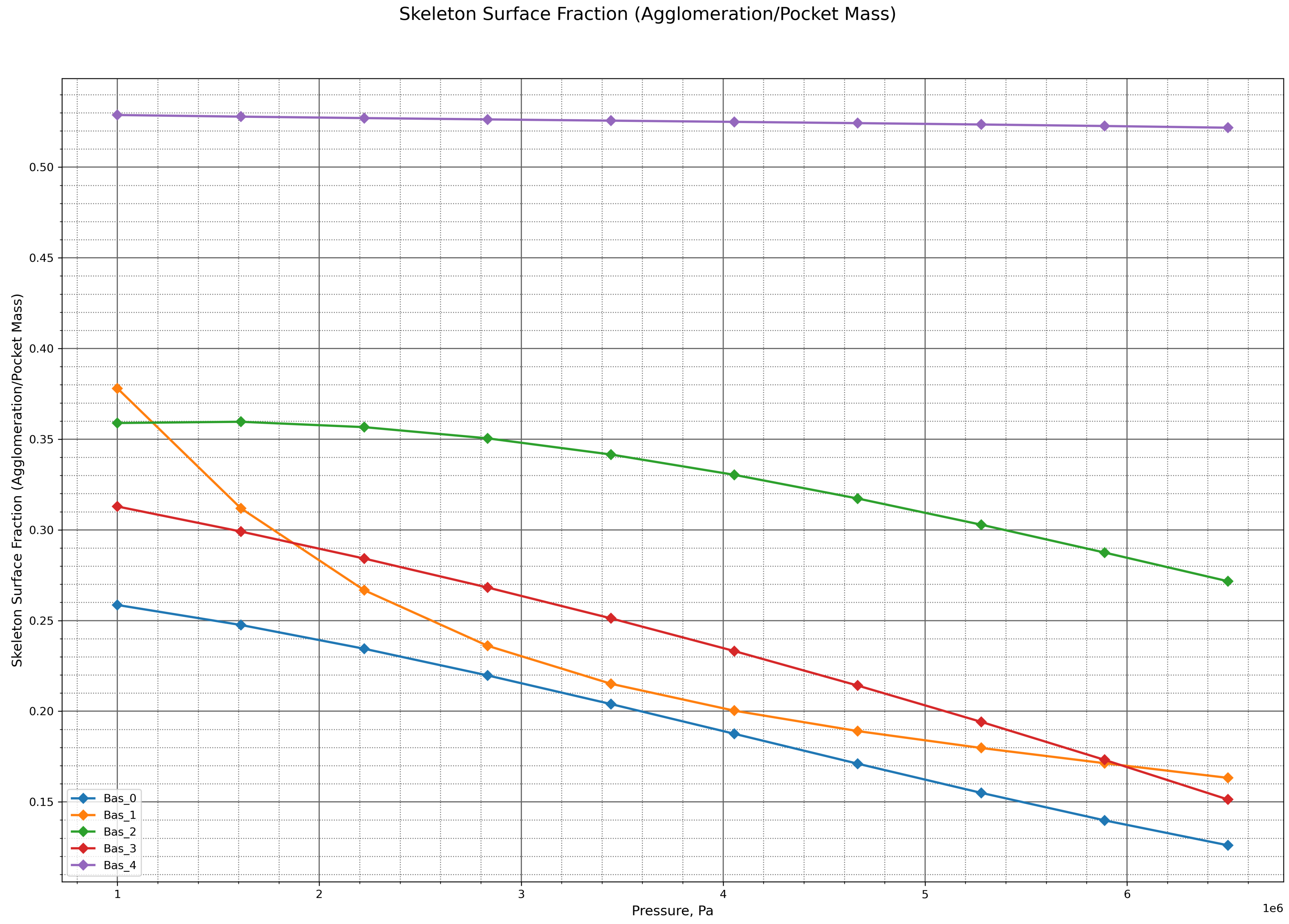


Constant Volume Heat Capacity (cᵥ), J/(kg·K)
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Constant Volume Heat Capacity (cᵥ), J/(kg·K)
Pressure, Pa
1e6
670
660
650
640
630
620
610
1
2
3
4
5
6


Temperature of the Flames, K
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Inter-Pocket Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Diffusion Flame (Pocket)
Temperature of the Flames, K
Pressure, Pa
1e6



Aluminum Agglomeration Fraction
Aluminum Agglomeration Fraction
Pressure, Pa
1e6
0.30
0.25
0.20
0.15
0.10
1
2
3
4
5
6
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4



Skeleton Surface Fraction (Agglomeration/Pocket Mass)
Skeleton Surface Fraction (Agglomeration/Pocket Mass)
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
1
2
3
4
5
6
1e6
Pressure, Pa
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4

Performance Report

System Information
Operating System: Unix 6.8.0.124 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.9 (.NET 10.0.9)
Architecture: X64 / X64

Total Execution Time
Total execution time: 08:59:14.302

Garbage Collection Information
Total Memory: 7749192 bytes
GC Pause Time Percentage: 0.15%
Generation 0 Collections: 39136
Generation 1 Collections: 3215
Generation 2 Collections: 322

Detailed Performance Metrics
Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 18811809
Maximum execution time: 36.4717 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.9761490590459145 ms
Execution time standard deviation: 0.14597594374831258 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 17834059
Maximum execution time: 25.4023 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.2832642618991492 ms
Execution time standard deviation: 0.13927367788735812 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 14559249
Maximum execution time: 35.5466 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.3374620187896109 ms
Execution time standard deviation: 0.06223223531891552 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 62.9801 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9739876041104358 ms
Execution time standard deviation: 0.14472823055215261 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 17737595
Maximum execution time: 27.4025 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28310513154688693 ms
Execution time standard deviation: 0.1386416630662587 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 14473392
Maximum execution time: 42.6986 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.33786809554385355 ms
Execution time standard deviation: 0.06231325454816525 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 57.0242 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9788717507821542 ms
Execution time standard deviation: 0.14167410960365348 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 17741962
Maximum execution time: 39.4188 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2830282420343167 ms
Execution time standard deviation: 0.13956038919279223 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 14455579
Maximum execution time: 31.3913 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33820491945013653 ms
Execution time standard deviation: 0.061201751100591326 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 38.1728 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.976860244163479 ms
Execution time standard deviation: 0.14178351562258218 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 17744302
Maximum execution time: 25.5093 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28281626657953046 ms
Execution time standard deviation: 0.13940472686263083 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 14453214
Maximum execution time: 28.24 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3379988341485666 ms
Execution time standard deviation: 0.05977793433868379 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 48.4077 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.978011987967175 ms
Execution time standard deviation: 0.14458666921639685 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 17743159
Maximum execution time: 50.3778 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28294271557271383 ms
Execution time standard deviation: 0.13981788541359041 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 14470622
Maximum execution time: 26.145 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.338075773667482 ms
Execution time standard deviation: 0.06016849993615002 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 33.5426 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9765754370064906 ms
Execution time standard deviation: 0.14005879340934013 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 17750234
Maximum execution time: 31.6386 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2837015569541046 ms
Execution time standard deviation: 0.13876791541057845 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 14491947
Maximum execution time: 32.31 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3390056122686426 ms
Execution time standard deviation: 0.060463594223285407 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 34.3468 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.975276416564551 ms
Execution time standard deviation: 0.1411688933263579 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 17741847
Maximum execution time: 63.3816 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2830482602121391 ms
Execution time standard deviation: 0.14004927098490114 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 14466778
Maximum execution time: 30.3734 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33786950622315914 ms
Execution time standard deviation: 0.06265085667312859 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 35.9725 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9779285645229665 ms
Execution time standard deviation: 0.14013834604197586 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 17754561

Maximum execution time: 26.0705 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28477373777930043 ms
Execution time standard deviation: 0.13824393745109373 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 14503332
Maximum execution time: 32.1627 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3411229506087988 ms
Execution time standard deviation: 0.06037039830914763 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 36.7425 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9753795338519874 ms
Execution time standard deviation: 0.14231823176493977 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 17737410
Maximum execution time: 42.0118 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28332454569749244 ms
Execution time standard deviation: 0.13927398280434516 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 14476180
Maximum execution time: 33.3733 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33794510539382633 ms
Execution time standard deviation: 0.06105057519704369 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 39.3509 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9773545455937009 ms
Execution time standard deviation: 0.14083390244691701 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 17748019
Maximum execution time: 40.3151 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28516187638180623 ms
Execution time standard deviation: 0.1398766640751696 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 14488426
Maximum execution time: 35.4061 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3389943037704653 ms
Execution time standard deviation: 0.06008554157977895 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 40.3612 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9741027694567721 ms

Execution time standard deviation: 0.1419369056863787 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 17745094
Maximum execution time: 27.589 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2834015172362665 ms
Execution time standard deviation: 0.13845839286773653 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 14504443
Maximum execution time: 29.1446 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33768271175257036 ms
Execution time standard deviation: 0.060162804219082475 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 34.8428 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9774636612897436 ms
Execution time standard deviation: 0.1406192649411593 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 17760119
Maximum execution time: 27.5467 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28422987363436736 ms
Execution time standard deviation: 0.1376791072558592 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 14522688
Maximum execution time: 33.4526 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3378001867629429 ms
Execution time standard deviation: 0.057571249527374566 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 47.9764 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9742530328202385 ms
Execution time standard deviation: 0.1446413484198856 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 17747723
Maximum execution time: 30.191 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28347587315855766 ms
Execution time standard deviation: 0.13858263063859655 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 14498234
Maximum execution time: 35.9573 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3376471492527232 ms
Execution time standard deviation: 0.06046372613986813 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 18711396
Maximum execution time: 40.2795 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9814022459254089 ms
Execution time standard deviation: 0.13848316432424332 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance
Call count: 17755112
Maximum execution time: 28.3032 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2849644856759989 ms
Execution time standard deviation: 0.13805415879924654 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance
Call count: 14548408
Maximum execution time: 35.2835 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33890509569153887 ms
Execution time standard deviation: 0.0596430476665852 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 36.1759 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9757467238788535 ms
Execution time standard deviation: 0.14186137598229856 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 17730227
Maximum execution time: 54.744 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28331947497915294 ms
Execution time standard deviation: 0.13937804269053697 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 14475088
Maximum execution time: 34.0417 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33838736222532534 ms
Execution time standard deviation: 0.06026029925285563 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 39.8065 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9767386437227568 ms
Execution time standard deviation: 0.13928795281299097 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 17740693
Maximum execution time: 44.8658 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2832172418799553 ms
Execution time standard deviation: 0.13890323087124637 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 14471367
Maximum execution time: 22.8925 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.33772858658067817 ms
Execution time standard deviation: 0.05999838647092629 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 36.0622 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.97340527254608 ms
Execution time standard deviation: 0.13801701797032578 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 17746122
Maximum execution time: 28.786 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2835870165661751 ms
Execution time standard deviation: 0.13677454347787896 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 14530544
Maximum execution time: 38.3234 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33793409016900455 ms
Execution time standard deviation: 0.05774340718034132 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 33.8144 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9779416630217811 ms
Execution time standard deviation: 0.14179626798296027 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 17736787
Maximum execution time: 29.9025 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2837869854105953 ms
Execution time standard deviation: 0.13930895879763155 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 14483542
Maximum execution time: 33.4083 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3382529389772364 ms
Execution time standard deviation: 0.06112358925599559 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 30.2477 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9746065762596595 ms
Execution time standard deviation: 0.13976438373253725 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 17754696
Maximum execution time: 37.3504 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28400153350975066 ms
Execution time standard deviation: 0.13785088205536875 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 14522290
Maximum execution time: 27.4699 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.339432379528254 ms
Execution time standard deviation: 0.0576068534807813 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 37.0969 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9732905867257836 ms
Execution time standard deviation: 0.14435282386897294 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 17735166
Maximum execution time: 35.0254 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2825713797547431 ms
Execution time standard deviation: 0.1392247759967301 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 14467901
Maximum execution time: 29.2669 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3372824021120572 ms
Execution time standard deviation: 0.06162324497357632 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 34.9922 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.975823527480113 ms
Execution time standard deviation: 0.1413777622281426 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 17740412
Maximum execution time: 26.5703 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2854463422664374 ms
Execution time standard deviation: 0.138853658485689 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 14507594
Maximum execution time: 31.5994 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3389257220804497 ms
Execution time standard deviation: 0.06014229864637545 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 53.4059 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9783910507373083 ms
Execution time standard deviation: 0.1390733164850939 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 17754624
Maximum execution time: 33.1549 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.2841137663799786 ms
Execution time standard deviation: 0.1380741717172013 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 14499639
Maximum execution time: 30.9278 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3390027143434586 ms
Execution time standard deviation: 0.059261408865170236 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 30.9867 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9795871425467426 ms
Execution time standard deviation: 0.14004523149866843 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 17731276
Maximum execution time: 43.7995 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28406132423294816 ms
Execution time standard deviation: 0.1392350124252154 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 14452820
Maximum execution time: 23.4734 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33822549660204465 ms
Execution time standard deviation: 0.0575983822714471 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 39.016 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9768116917519692 ms
Execution time standard deviation: 0.1397392298531028 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 17760668
Maximum execution time: 28.3988 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2841191315213958 ms
Execution time standard deviation: 0.13755853438225418 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 14525552
Maximum execution time: 31.2865 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3383105965543039 ms
Execution time standard deviation: 0.06086381166490137 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 45.6128 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9762037613708188 ms
Execution time standard deviation: 0.1421231233109268 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 17722357
Maximum execution time: 41.4094 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2816391839019766 ms
Execution time standard deviation: 0.139622875521288 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 14418188
Maximum execution time: 37.5141 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33760012702012754 ms
Execution time standard deviation: 0.06098421919728793 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 42.0056 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9749805265358157 ms
Execution time standard deviation: 0.14161809297557282 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 17737626
Maximum execution time: 32.3568 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28290524892110824 ms
Execution time standard deviation: 0.13880851006690176 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 14453987
Maximum execution time: 25.8471 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3382527430113673 ms
Execution time standard deviation: 0.05956750148840937 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 30.155 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.977281013004034 ms
Execution time standard deviation: 0.1397124832538504 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 17742907
Maximum execution time: 35.7698 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2832740514617664 ms
Execution time standard deviation: 0.13825175802546769 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 14483414
Maximum execution time: 36.3947 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.338077407336459 ms
Execution time standard deviation: 0.059025115728162304 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 18711396

Maximum execution time: 40.0003 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9771228537144406 ms
Execution time standard deviation: 0.13869775774498092 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 17736394
Maximum execution time: 39.5886 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.283689430010314 ms
Execution time standard deviation: 0.1380327591670678 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 14482604
Maximum execution time: 28.5888 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33914147866637007 ms
Execution time standard deviation: 0.05689039481035382 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 36.0176 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9773337750428194 ms
Execution time standard deviation: 0.14198082835270698 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 17757739
Maximum execution time: 29.9363 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2835821633936368 ms
Execution time standard deviation: 0.13844784306517255 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 14500419
Maximum execution time: 26.1933 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33874181252969754 ms
Execution time standard deviation: 0.05945045708843202 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 32.1653 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9770110637336512 ms
Execution time standard deviation: 0.13780818557450392 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 17733985
Maximum execution time: 46.424 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28466232344281484 ms
Execution time standard deviation: 0.1386821678854038 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 14483841
Maximum execution time: 42.377 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3397845832676172 ms

Execution time standard deviation: 0.05866404544767016 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 39.5936 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9766444553682444 ms
Execution time standard deviation: 0.1383580805073851 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 17740327
Maximum execution time: 63.4053 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28354058470281435 ms
Execution time standard deviation: 0.1396394539969869 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 14451858
Maximum execution time: 27.8151 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3388957982911094 ms
Execution time standard deviation: 0.05797707696349522 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 29.9581 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.979057010957384 ms
Execution time standard deviation: 0.13584458286047682 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 17746640
Maximum execution time: 32.7769 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28523456786750684 ms
Execution time standard deviation: 0.13823289815929177 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 14483158
Maximum execution time: 30.2488 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3391962099426114 ms
Execution time standard deviation: 0.05738017265769665 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 57.9686 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9743283124412484 ms
Execution time standard deviation: 0.1428676472909579 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance
Call count: 17755914
Maximum execution time: 47.6495 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28252497009736754 ms
Execution time standard deviation: 0.13862192485049657 ms

Method: Telemetry/Proc/32/OptimizationProblemExecutionPerformance

Call count: 14484379
Maximum execution time: 49.4171 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.33760460632108213 ms
Execution time standard deviation: 0.06268078225355948 ms

Method: Telemetry/Proc/33/OptimizationProblemExecutionPerformance
Call count: 18711396
Maximum execution time: 33.2261 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.9788729727274935 ms
Execution time standard deviation: 0.14412918517999154 ms

Method: Telemetry/Proc/33/OptimizationProblemExecutionPerformance
Call count: 17744538
Maximum execution time: 28.3838 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28412106094845024 ms
Execution time standard deviation: 0.13803726368190533 ms

Method: Telemetry/Proc/33/OptimizationProblemExecutionPerformance
Call count: 14526471
Maximum execution time: 22.9701 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3376331047093058 ms
Execution time standard deviation: 0.06027374492258645 ms